

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- MCQ

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO3	Assessment:	ASSESSMENT TOOL 2- MCQ
Weightage:	20%	Total Marks:	50
CO3 Statement:	Superscalar Processing: Out-of-order execution, Speculative execution, Dispatch and execution units. Advanced Features: Hyper-threading and simultaneous multithreading (SMT). (BL3)		
Student Name:		Reg.No.:	
Department:		Date:	

ANNEXURE A
MCQ Questions

1. A 4-issue superscalar processor can ideally issue how many instructions per clock cycle?

- A) 1
- B) 2
- C) 4
- D) 8

2. A processor has four execution units, but only two independent instructions are available in a cycle. What is the maximum number of instructions that can be executed in that cycle?

- A) 1
- B) 2
- C) 4
- D) 8

3. Which situation most directly limits the performance of a superscalar processor?

- A) High instruction-level parallelism
- B) Independent instructions
- C) Strong data dependencies
- D) Multiple execution units

4. Consider the instructions: I1: $R1 \leftarrow R2 + R3$ and I2: $R4 \leftarrow R5 + R6$. If sufficient execution units are available, what can happen?

- A) I2 must wait for I1
- B) Both can execute concurrently
- C) I1 must wait for I2
- D) Neither can execute

5. A 4-wide superscalar processor executes an average of 2 instructions per cycle. What is its effective IPC?

- A) 1
- B) 2
- C) 4
- D) 8

6. If a processor's issue width is increased from 2 to 4 but the available instruction-level parallelism remains low, what is the likely result?

- A) Performance always doubles
- B) Performance may improve only slightly
- C) Performance becomes zero
- D) Clock frequency must double

7. Which mechanism dynamically identifies instructions that are ready for execution?

- A) Scheduler
- B) Hard disk controller
- C) DMA controller
- D) Memory decoder

8. Why are multiple execution units important in superscalar processors?

- A) They allow independent operations to proceed simultaneously
- B) They eliminate the need for instructions
- C) They replace main memory
- D) They reduce program size

9. Consider: I1: $R1 \leftarrow R2 + R3$ and I2: $R4 \leftarrow R1 \times R5$. Why can I2 not execute before I1 produces R1?

- A) Control dependency
- B) RAW dependency
- C) WAW dependency
- D) Structural dependency

10. Which technique allows a processor to exploit instructions that are independent even when earlier instructions are delayed?

- A) Out-of-order execution
- B) Sequential execution
- C) Memory mapping
- D) Paging

11. Instruction I1 is waiting for memory data, while I2 is independent and ready. What can an out-of-order processor do?

- A) Stall I2 until I1 finishes
- B) Execute I2 while I1 waits
- C) Delete I1
- D) Restart the processor

12. What is the major advantage of out-of-order execution?

- A) It keeps execution units busy
- B) It eliminates all memory accesses
- C) It reduces the number of registers
- D) It eliminates branch instructions

13. Consider the sequence: I1: $R1 \leftarrow R2 + R3$, I2: $R4 \leftarrow R1 + R5$, I3: $R6 \leftarrow R7 + R8$. If I1 is delayed, which instruction can potentially execute first?

- A) I2 only
- B) I3 only
- C) I1 only
- D) I1 and I2 only

14. Why is register renaming used in an out-of-order processor?

- A) To eliminate false register dependencies
- B) To increase instruction length
- C) To reduce cache size
- D) To remove the ALU

15. Two instructions write to the same architectural register but have no true data dependency. Which hazard can occur?

- A) RAW
- B) WAR
- C) WAW
- D) Cache hazard

16. Register renaming primarily converts architectural registers into:

- A) Physical registers
- B) Cache blocks
- C) Memory pages
- D) Instruction codes

17. Which dependency represents a true data dependency?

- A) RAW
- B) WAR
- C) WAW
- D) None

18. If instruction I2 reads a register before instruction I1 writes a new value to that register, the dependency is:

- A) RAW
- B) WAR
- C) WAW
- D) Structural

19. Which structure helps an out-of-order processor preserve the original program order when committing results?

- A) Reorder Buffer
- B) ALU
- C) Branch Predictor
- D) Cache Controller

20. Why should results generally be committed in program order?

- A) To maintain precise architectural state
- B) To increase instruction length
- C) To reduce register count
- D) To disable speculation

21. An instruction has completed execution but cannot yet update the architectural state because an older instruction has not completed. Which mechanism can handle this situation?

- A) Reorder Buffer
- B) RAM refresh
- C) DMA
- D) I/O controller

22. Which situation is most suitable for out-of-order execution?

- A) Independent instructions with different execution delays
- B) A sequence where every instruction depends on the previous one
- C) A processor with no execution units
- D) A program containing no instructions

23. Why does a processor use speculative execution?

- A) To continue useful execution before branch outcomes are known
- B) To eliminate registers
- C) To reduce memory capacity
- D) To stop pipeline execution

24. A branch predictor predicts that a conditional branch will be taken. What does the processor typically do?

- A) Fetch and possibly execute instructions from the predicted path
- B) Stop fetching instructions permanently
- C) Delete the branch instruction
- D) Reset the processor

25. What happens when speculative execution follows an incorrect branch prediction?

- A) Incorrect speculative work is discarded
- B) All previous programs are deleted
- C) The processor permanently stops
- D) The cache is destroyed

26. A pipeline has a branch misprediction penalty of 5 cycles. If 10 branch mispredictions occur, how many cycles are lost, assuming no overlap?

- A) 5 cycles
- B) 10 cycles
- C) 50 cycles
- D) 100 cycles

27. Which factor has the greatest impact on the effectiveness of speculative execution?

- A) Branch prediction accuracy
- B) Keyboard speed
- C) Monitor resolution
- D) Disk capacity

28. If branch prediction accuracy decreases significantly, what is the likely effect?

- A) More pipeline flushes and reduced performance
- B) Increased memory capacity
- C) Elimination of all hazards
- D) Increased instruction length

29. Which prediction would generally be most beneficial for a branch executed 100 times and taken 95 times?

- A) Predict taken
- B) Predict not taken

- C) Predict randomly
- D) Never predict

30. Speculative instructions must not permanently change the architectural state until:

- A) Their correctness is confirmed
- B) They enter the cache
- C) The program terminates
- D) The clock stops

31. Which mechanism helps discard the results of incorrectly speculated instructions?

- A) Recovery mechanism
- B) Cache replacement
- C) DMA controller
- D) Memory decoder

32. What is the major performance benefit of accurate branch prediction?

- A) Reduced control-flow stalls
- B) Reduced RAM size
- C) Reduced instruction count
- D) Elimination of registers

33. An instruction requiring floating-point addition is dispatched to which type of execution unit?

- A) Floating-Point Unit
- B) Integer ALU
- C) Branch Unit
- D) Load/Store Unit

34. A load instruction generally requires which execution resource?

- A) Load/Store Unit
- B) Floating-Point Unit
- C) Branch Predictor
- D) Integer Register only

35. If all integer ALUs are busy but a floating-point unit is free, what is the likely result for a ready integer instruction?

- A) It waits for an appropriate integer execution unit
- B) It executes on the floating-point unit automatically
- C) It is deleted
- D) It becomes a branch instruction

36. What is the primary role of the dispatch stage?

- A) Send instructions toward appropriate execution resources
- B) Store the operating system
- C) Perform disk operations
- D) Generate program source code

37. A processor has two integer ALUs and one floating-point unit. Which instruction mix can best utilize all three simultaneously?

- A) Two integer operations and one floating-point operation
- B) Three floating-point operations
- C) Three branch instructions only
- D) One integer operation only

38. If execution units are frequently idle because instructions depend on previous results, the main limitation is:

- A) Lack of instruction-level parallelism
- B) Excessive memory capacity
- C) Excessive register count
- D) High display resolution

39. A physical CPU core supports two hardware threads using SMT. How many logical processors can the operating system typically see from that core?

- A) 1
- B) 2
- C) 4
- D) 8

40. Two threads execute on the same SMT core. Which statement is correct?

- A) They can share the core's execution resources
- B) Each thread receives a completely separate physical core
- C) They cannot execute simultaneously
- D) One thread must always remain inactive

41. SMT is most useful when:

- A) One thread leaves some processor resources underutilized
- B) The processor has no execution units
- C) Only one instruction exists
- D) The system has no memory

42. Which statement best distinguishes SMT from multicore processing?

- A) SMT allows multiple threads to share one physical core
- B) SMT requires one physical core per thread
- C) Multicore uses only one execution stream
- D) Multicore eliminates parallelism

43. A single-threaded workload completely occupies all execution resources of a core. What is the likely benefit of enabling SMT for that workload?

- A) Limited performance improvement
- B) Guaranteed 2× performance
- C) Guaranteed 4× performance
- D) Performance becomes zero

44. Two threads on the same SMT core compete heavily for the same execution resources. What is the likely result?

- A) Performance may not scale linearly
- B) Performance always doubles
- C) Both threads stop permanently
- D) Memory capacity doubles

45. Which resources may be shared by multiple SMT threads?

- A) Execution units and caches
- B) Separate physical cores only
- C) Separate processors only
- D) Independent motherboards

46. Why does SMT generally provide less performance improvement than adding a completely independent physical core?

- A) Threads share hardware resources within the same core
- B) SMT uses no registers
- C) SMT disables pipelining
- D) SMT eliminates execution units

47. A processor has 4 physical cores and each core supports 2-way SMT. How many logical processors can the operating system typically detect?

- A) 4
- B) 6
- C) 8
- D) 16

48. Which workload is most likely to benefit from SMT?

- A) Multiple threads with complementary resource demands
- B) A single instruction with no parallelism
- C) A program that never executes
- D) A workload with completely unavailable memory

49. Which combination best represents techniques used to exploit instruction and thread-level parallelism?

- A) Superscalar execution and SMT
- B) Paging and segmentation
- C) DMA and interrupt handling
- D) Cache replacement and disk formatting

50. A processor combines superscalar execution, out-of-order execution, speculative execution, and SMT. What is the primary overall goal?

- A) Maximize utilization of processor resources and improve throughput
- B) Reduce the number of instructions in a program to zero
- C) Eliminate the need for main memory
- D) Replace the operating system with hardware

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I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Problem Understanding (20%)	Method Selection (20%)	Solution Process (25%)	Accuracy of Calculation (20%)	Interpretation of Results (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation